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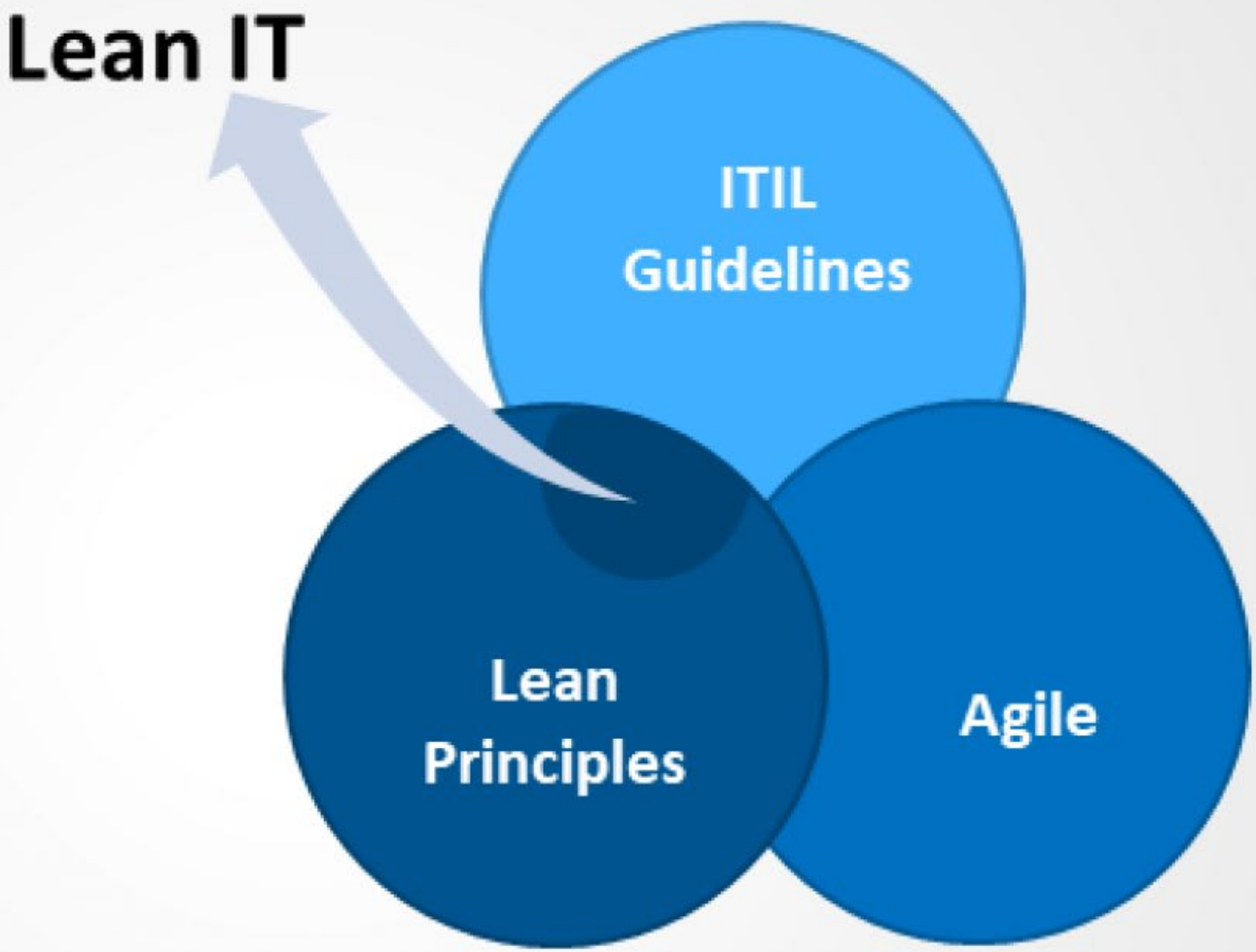
IT Service Management (ITSM)

Agila and Lean IT(SM) 2024


Lean IT
ITIL
Guidelines
Lean
Principles
Agile

The agile manifesto

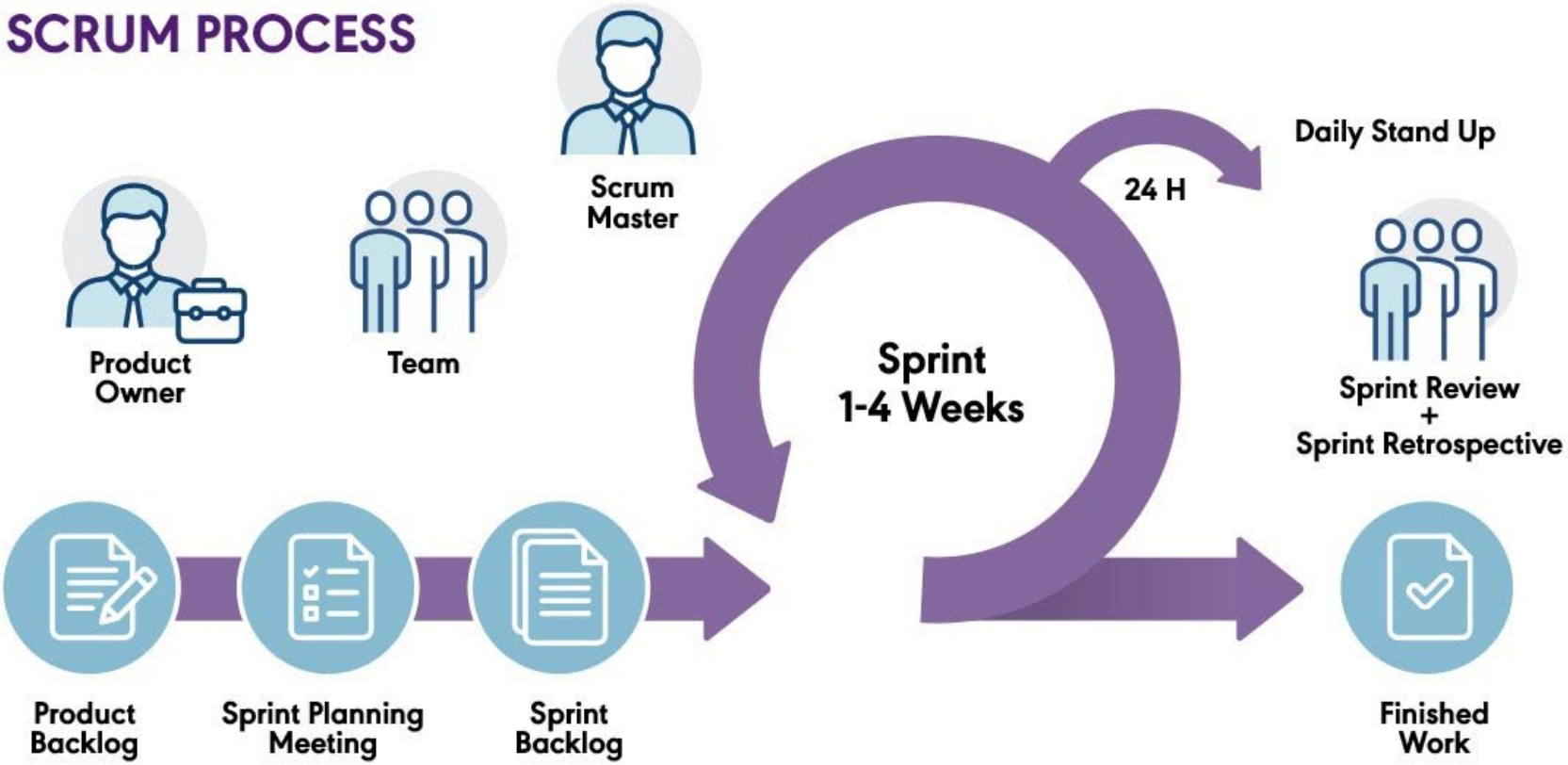
We are uncovering better ways of developing
digital services by doing it and helping others do it. Through this work
we have come to value:

The Agile Manifesto

Individuals and interactions	over	Processes and Tools
Working Product	over	Comprehensive Documentation
Customer Collaboration	over	Contract Negotiation
Responding to change	over	Following a plan

That is, while there is value in the items on the right, we value the items on the left more.

www.agilemanifesto.org


SCRUM PROCESS
Scrum Master
Product Owner
Team
Daily Stand Up
24 H
Sprint 1-4 Weeks
Sprint Review + Sprint Retrospective
Product Backlog
Sprint Planning Meeting
Sprint Backlog
Finished Work

Lean ITSM

AN ITSM PERSPECTIVE

JÖNKÖPING UNIVERSITY
School of Engineering

Origins of Lean

- Lean development at Toyota is called **Toyota Production System (TPS)**
  - Japan has few natural resources
  - Japan had to import material (and resources)
  - Japan had to be effective and efficient to be able to compete

- Taiichi Ohno
  - Inventor of TPS and Just-In-Time manufacturing
  - "Costs do not exist to be calculated. Costs exist to be reduced." (Taiichi Ohno, 1988)
  - Inspired by **Plan - Do - Check - Act** (PDCA – cycle or Demings cycle). Consequently, Lean IT is related to Continous Service Improvement

What is Lean?

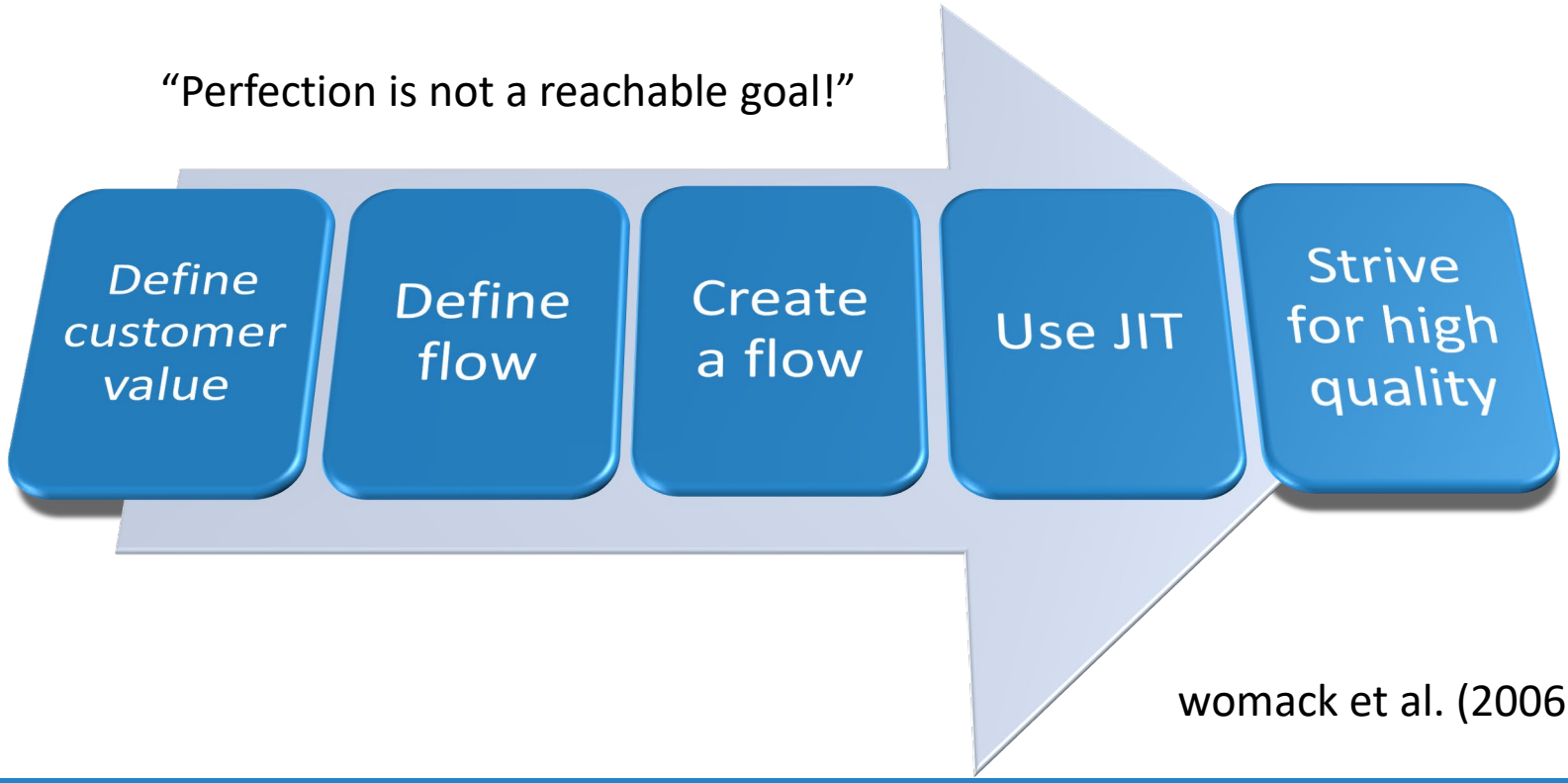
- Lean has many definitions and researchers and practitioners continuously adds new tools and concepts into the TPS concept and call it Lean...

- **Goal**: Reduce waste in a system and enable greater value for the customer through:
  - Philosophy (Kaizen, improvement culture, sharing etc.)
  - Tools (5S, 5 Whys, Muda, Kanban etc. ...)

- Compare with key attributes of IT Service Management…

Lean Process?

"Perfection is not a reachable goal!"



womack et al. (2006)

TSP vs LEAN

TPS **is not** Lean!

”*Triumph of the lean production system*” (Krafcik,1988)

“The machine that changed the world” (Womack et al., 1990)

Basics of Lean IT

- Enable value for the customer
- Process mappings and value streams
- Smooth/ continuous flow
- Optimization of processes
- Pulling system
- Involvement

Seven Principles of Lean IT

1. Eliminate Waste
2. Amplify Learning
3. Decide as late as possible
4. Deliver as fast as possible
5. Empower the team
6. Build integrity inn
7. See the whole

Principle 1: Eliminate Waste

- Remove waste that adds no value to project

- Three types of waste:
  - Waste in service (software/hardware) development
  - Waste in project management
  - Waste in workforce potential

Eliminate Waste: Code & Infrastructure

- Partially completed work
  - The work can become outdated and unusable if we wait to long
  - Solution: Iterative cycle with modular code
- Defects
  - Correction and retesting
  - Solution: Up-to-date test suite, customer feedback

Eliminate Waste: Project Management

- Redundant/unnecessary processes and documentation
  - Solution: Documentation review
  - But does it mean that documentation is unnecessary?
- Code Handoffs
  - Loss of knowledge
  - Solution: Don’t handoff code
- Extra Functions/Features
  - The customer doesn’t want or need a feature
  - Solution: Continuous interaction with customer

Eliminate waste: Workforce Potential

- Task Switching
  - The danger of multi-tasking
  - Solution: Focus on a task each release
  - Example (tool): https://cirillocompany.de/pages/pomodoro-technique
- Waiting for instructions or information
  - Developers are too expensive to be sitting around!
  - Solution: Allow developers to make decisions, access to information

Principle 2: Amplify Learning

- Learning is not compulsory... neither is survival.
- Full knowledge is impossible, need to learn as project is developed.
- The customer should be learning too!
- Use a “scientific” method
  - Create hypothesis
  - Conduct rapid tests of possible solutions/prototypes
  - Compare results
  - Implement the best solution

Principle 3: Delay Commitment

- Decisions are often made on incomplete or incorrect information
- Delay until the last responsible moment
- Benefits:
  - More knowledge for decision making
  - Leaves options open

Principle 4: Empower the Team

- Make decisions as “low” as possible

- Shift from complete documentation to goals and guidelines
  - Developers closest to code know the problem best
  - Developers can get feedback for personal improvement
  - Developers must be experienced in domain

- “The best executive is one who has sense enough to pick good people to do what he wants done, and self-restraint enough to keep from meddling with them while they do it.” (Theodore Roosevelt)

Principle 5: Deliver Fast

- Customers change their mind.
  - Deliver so fast so they don't have time to! (How does it relate to S-D Logic?)
- Short release cycles
  - Two-four weeks?
- Requirements:
  - Don't overload developers with requests
  - Remember - No partially completed work

Principle 6: Build Integrity In

- Perceived Integrity
  - Does exactly what the customer wants, even if they don't know it yet
    - **What is the problem with this statement?**
  - Again, continual interaction with the customer

- Conceptual Integrity
  - The product/service works smoothly and functions well
  - The product/service parts works together

Principle 7: See the Whole

- Optimize the entire service, not just parts
- Risk: short delivery cycles can lead to optimized parts and sub optimized whole
- Don’t have customer overestimate what the product will do

Kaizen

- Japanese word that means to make many **small** changes for the better
  - Kai = change
  - Zen = good

- Improvements are best when they are created by the person doing the work

- The person doing the work uses their own common sense and intuition

- Tool ”Go & see”

Lean Practice - Discipline of the 5's

- Sort (when in doubt, throw it out)
- Straighten (labeling, color coding, filing)
- Shine (keep cleaning supplies handy, replace worn, torn, items not working)
- Standardize (create layouts, write checklists)
- Sustain (create a new work habit, check it regularly – at least once a week)

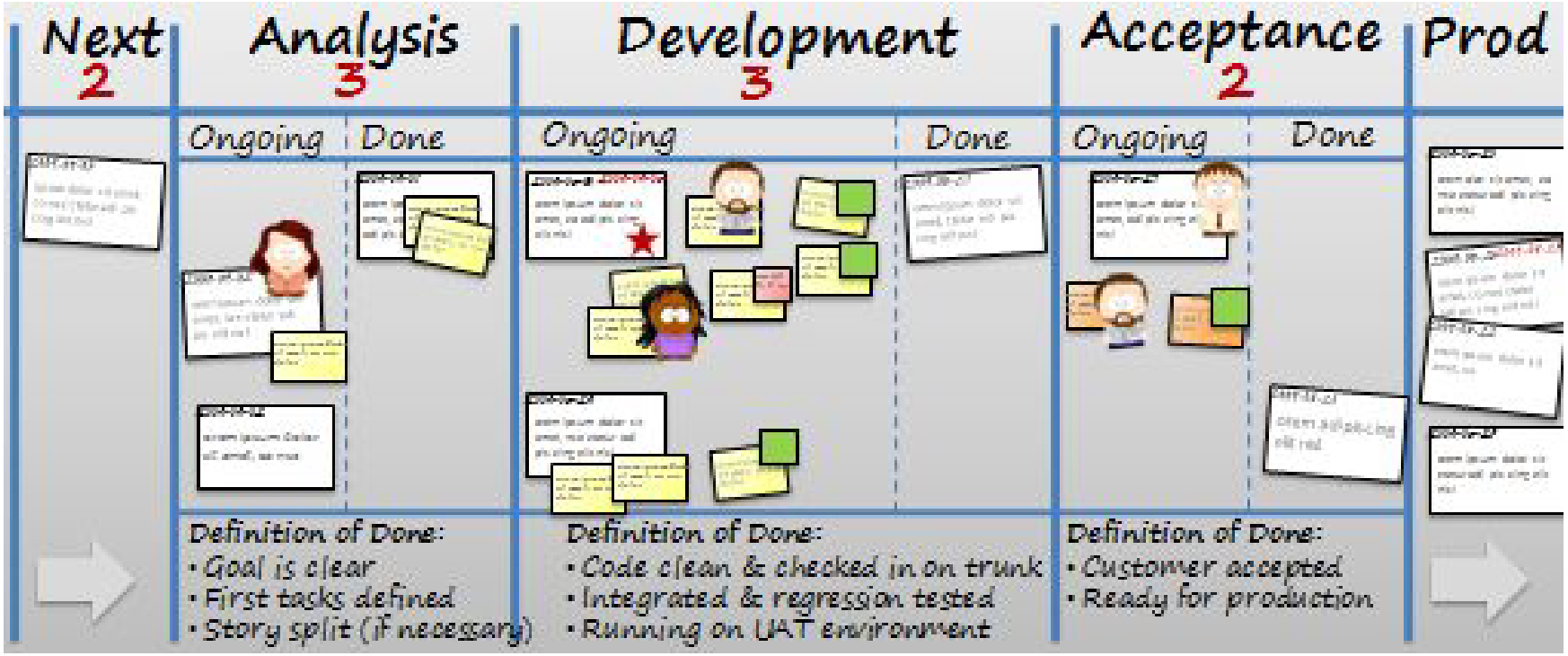
Kanban

看板 – Kanban limit WIP

- Kanban: Kan & Ban - means “visual card,” “signboard,” or “billboard.”
- Is a tool/component of Lean supporting just-in-time (JiT)
- Toyota originally used Kanban cards to limit the amount of inventory tied up in “work in progress” on a manufacturing floor
- Not only is excess inventory waste, time spent producing it is time that could be used elsewhere
- Kanban cards act as a form of “currency” representing how WIP is allowed in a system.

Kanban ”rules” supporting JIT

1. Visualization
2. Work in progress (WIP) using pull
3. Optimization and measurements



https://www.crisp.se/gratis-material-och-guider/kanban

Why kanban in ITSM

Using a Kanban approach in ITSM drops time-boxed iterations (e.g., Scrum) in favor of focusing on a *continuous flow*.